Full Mock Test 2 · Paper 1

# Question 1

What is the value of

$$\left(\left((2+1)^{-1}+1\right)^{-1}+1\right)^{-1}+1$$

A $\frac{5}{8}$

B $\frac{11}{7}$

C $\frac{8}{5}$

D $\frac{18}{11}$

E $\frac{15}{8}$

Full Mock Test 2 · Paper 1

# Question 2

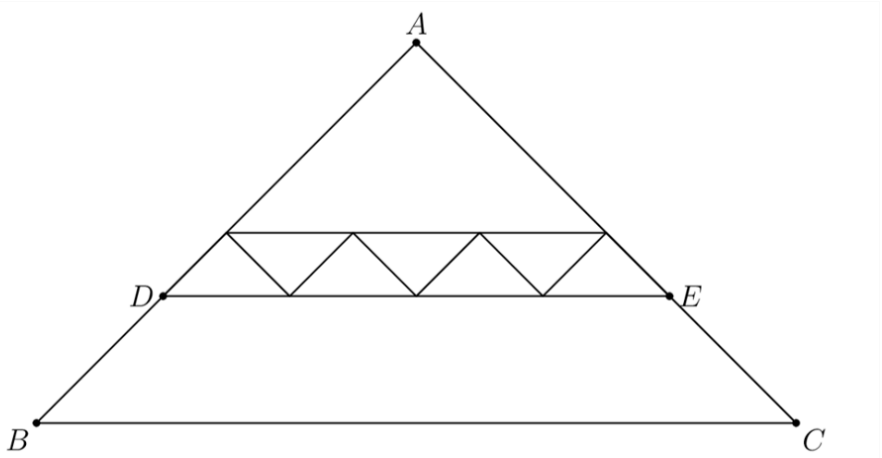
For how many (not necessarily positive) integer values of $n$ is the value of $4000 \cdot \left(\frac{2}{5}\right)^n$ an integer?

A $4$
B $5$
C $6$
D $7$
E $8$
F $9$
G $10$
H $11$

Full Mock Test 2 · Paper 1

# Question 3

All of the triangles in the diagram below are similar to isosceles triangle $\triangle ABC$, in which $AB = AC$. Each of the 7 smallest triangles has area $1$, and $\triangle ABC$ has area $40$. What is the area of trapezoid $DBCE$?




A $16$

B $18$

C $20$

D $22$

E $24$

Full Mock Test 2 · Paper 1

# Question 4

Suppose that real number $x$ satisfies

$$\sqrt{49 - x^2} - \sqrt{25 - x^2} = 3$$

What is the value of $\sqrt{49 - x^2} + \sqrt{25 - x^2}$?

A $8$

B $\sqrt{33} + 8$

C $9$

D $2\sqrt{10} + 4$

E $12$

Full Mock Test 2 · Paper 1

# Question 5

Two open intervals (−1, 3) and (k − 1, k + 1) overlap in an interval of length exactly 1. What is the sum of the squares of all possible real values of k?

**A** 10

**B** 4

**C** 5

**D** 8

**E** 2

Full Mock Test 2 · Paper 1

# Question 6

Two upright cylindrical vessels have internal radii 2r and 3r. Initially the water in each is h deep. Water is poured from the narrower vessel into the wider one until the depth in the wider vessel is twice that in the narrower one. What fraction of the narrower vessel's original water is poured out?

**A** 1/3

**B** 4/11

**C** 9/22

**D** 1/2

**E** 9/13

Full Mock Test 2 · Paper 1

# Question 7

A convergent geometric progression has a positive first term and a negative second term. Its sum from the third term onwards is one quarter of its total sum. Its third and fourth terms add to 3. What is the total sum?

**A** 8

**B** 12

**C** 24

**D** 18

**E** 16

Full Mock Test 2 · Paper 1

# Question 8

How many ordered pairs of real numbers $(x, y)$ satisfy the following system of equations?

$$x + 3y = 3$$

$$\big||x| - |y|\big| = 1$$

A $1$

B $2$

C $3$

D $4$

E $5$

F $6$

G $7$

H $8$

Full Mock Test 2 · Paper 1

# Question 9

The coefficient of $x^2$ in a quadratic f(x) is 1, and its two zeros are 2 units apart. Its graph meets the reflected graph $y = f(4 - x)$ at (2, 8). What is the sum of all possible values of f(0)?

**A** 12

**B** 16

**C** 24

**D** 20

**E** 32

Full Mock Test 2 · Paper 1

# Question 10

A circle passes through (0, 0), (8, 0) and B = (0, b), where b > 0. The tangent at B passes through (3, 10). Two values of b are possible. What is their difference?

**A** 1

**B** 4

**C** 3

**D** 2

**E** 6

Full Mock Test 2 · Paper 1

# Question 11

A real number $a$ is chosen so that the coefficients of $x^3$ and $x^5$ in $(1 + x)^6(1 - ax^2)$ are equal. What is the coefficient of $x^4$ in that product?

**A** 0

**B** 30

**C** 15

**D** 28

**E** 14

Full Mock Test 2 · Paper 1

# Question 12

A point at angle x moves once round the unit circle, with $0 \le x < 2\pi$. For what fraction of the full turn is the point above the horizontal axis while $\cos(2x) > \cos x$?

**A** 1/12

**B** 1/4

**C** 1/6

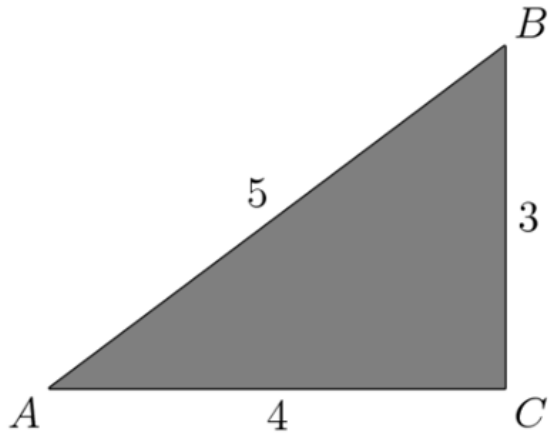
**D** 1/3

**E** 1/2

Full Mock Test 2 · Paper 1

# Question 13

A paper triangle with sides of lengths $3$, $4$, and $5$ inches, as shown, is folded so that point $A$ falls on point $B$. What is the length in inches of the crease?




A $1+\frac{1}{2}\sqrt{2}$

B $\sqrt{3}$

C $\frac{7}{4}$

D $\frac{15}{8}$

E $2$

Full Mock Test 2 · Paper 1

# Question 14

Real numbers a and b satisfy $2^a + 2^b = 12$ and $2^{(a+b)} = 32$. What is the value of $2^{(a-b)} + 2^{(b-a)}$?

**A** 3/2

**B** 5/2

**C** 9/4

**D** 2

**E** 4

Full Mock Test 2 · Paper 1

# Question 15

A rectangular plot has sides parallel to the coordinate axes. Its upper-right corner is (6, 5). Its lower-left corner (u, v) is in the first quadrant on the line u + 2v = 8. What is the greatest possible area of the plot?

**A** 8

**B** 7

**C** 6

**D** 9

**E** 10

Full Mock Test 2 · Paper 1

# Question 16

For a fixed real number c, both inequalities $2^{x-c} \le M$ and $4^{c-x} \le M$ must hold for every x in [−1, 5]. What is the smallest M obtainable by choosing c?

**A** 4

**B** 8

**C** 32

**D** 16

**E** 64

Full Mock Test 2 · Paper 1

# Question 17

Which of the following describes the set of values of $a$ for which the curves $x^2 + y^2 = a^2$ and $y = x^2 - a$ in the real $xy$-plane intersect at exactly 3 points?

A $a = \frac{1}{4}$

B $\frac{1}{4} < a < \frac{1}{2}$

C $a > \frac{1}{4}$

D $a = \frac{1}{2}$

E $a > \frac{1}{2}$

Full Mock Test 2 · Paper 1

# Question 18

A function $f$ is defined recursively by

$$f(1) = f(2) = 1 \quad \text{and} \quad f(n) = f(n-1) - f(n-2) + n$$

for all integers $n \geq 3$. What is $f(2018)$?

A $2016$

B $2017$

C $2018$

D $2019$

E $2020$

Full Mock Test 2 · Paper 1

# Question 19

In the $xy$-plane, a parabola has vertex $(9, -14)$ and intersects the $x$-axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where $a$, $b$, and $c$ are constants, which of the following could be the value of $a + b + c$?

A $-23$

B $-19$

C $-14$

D $-12$

Full Mock Test 2 · Paper 1

# Question 20

What is

$$\sum_{i=1}^{100}\sum_{j=1}^{100}(i+j)?$$

A $100{,}100$

B $500{,}500$

C $505{,}000$

D $1{,}001{,}000$

E $1{,}010{,}000$

Full Mock Test 2 · Paper 2

# Question 1

The sum of two nonzero real numbers is 4 times their product. What is the sum of the reciprocals of the two numbers?

A 1

B 2

C 4

D 8

E 12

Full Mock Test 2 · Paper 2

# Question 2

Alice, Bob, and Charlie were on a hike and were wondering how far away the nearest town was. When Alice said, "We are at least $6$ miles away," Bob replied, "We are at most $5$ miles away." Charlie then remarked, "Actually the nearest town is at most $4$ miles away." It turned out that none of the three statements were true. Let $d$ be the distance in miles to the nearest town. Which of the following intervals is the set of all possible values of $d$?

A $(0, 4)$
B $(4, 5)$
C $(4, 6)$
D $(5, 6)$
E $(5, \infty)$

Full Mock Test 2 · Paper 2

# Question 3

Ms. Carroll promised that anyone who got all the multiple choice questions right on the upcoming exam would receive an A on the exam. Which one of these statements necessarily follows logically?

**A** If Lewis did not receive an A, then he got all of the multiple choice questions wrong.

**B** If Lewis did not receive an A, then he got at least one of the multiple choice questions wrong.

**C** If Lewis got at least one of the multiple choice questions wrong, then he did not receive an A.

**D** If Lewis received an A, then he got all of the multiple choice questions right.

**E** If Lewis received an A, then he got at least one of the multiple choice questions right.

Full Mock Test 2 · Paper 2

# Question 4

At a gathering of $30$ people, there are $20$ people who all know each other and $10$ people who know no one. People who know each other hug, and people who do not know each other shake hands. How many handshakes occur?

A $240$

B $245$

C $290$

D $480$

E $490$

Full Mock Test 2 · Paper 2

# Question 5

For an integer $n \geq 4$, a student compares the coefficient of $x^4$ in $(1 + x)^n$ with twice the coefficient of $x^2$ in $(1 + x)^{n-1}$. What is the least $n$ for which the first quantity is strictly larger?

**A** 5

**B** 6

**C** 8

**D** 7

**E** 9

Full Mock Test 2 · Paper 2

# Question 6

A student claims: ‘If 6 divides n(n + 1), then 6 divides at least one of n and n + 1.’ How many integers n from 1 to 60 inclusive are counterexamples?

**A** 10

**B** 15

**C** 50

**D** 30

**E** 40

**F** 45

**G** 20

Full Mock Test 2 · Paper 2

# Question 7

A trapezium-rule estimate for the integral of a quadratic $ax^2 + bx + c$, with $a > 0$, uses just two strips: [0, 1] and [1, 3]. Halving only the first strip reduces the estimate by D. By how much would halving only the second strip reduce the original estimate?

**A** 2D

**B** 8D

**C** 4D

**D** 6D

**E** 3D

Full Mock Test 2 · Paper 2

# Question 8

Real numbers u < v are fixed. For every straight-line function f(x) = mx + c, the integral of f from u to v equals (v − u)f(1). Which condition is necessary and sufficient for this property?

**A** u + v = 2

**B** u ≤ 1 ≤ v

**C** uv = 1

**D** v − u = 2

**E** u = 0 and v = 2

Full Mock Test 2 · Paper 2

# Question 9

P(n) is a statement about positive integers. You know P(4) is true, and whenever P(n) is true, both P(n + 4) and P(n + 6) are true. What is the greatest even integer at least 4 for which these facts do not force P(n) to be true?

**A** 12

**B** 8

**C** 10

**D** 6

**E** There is no greatest such integer

Full Mock Test 2 · Paper 2

# Question 10

For $x > 1$ put $a = \log_2 x$ and $b = \log_x 8$. Which value of k makes the equation $a + b = k$ have exactly one possible value of x?

**A** $\sqrt{3}$

**B** 3

**C** $2\sqrt{3}$

**D** 4

**E** 6

Full Mock Test 2 · Paper 2

# Question 11

An integer $d > 0$ divides $n(n + 1)(2n + 1)$ for every positive integer $n$. For how many different values of $d$ is this possible?

**A** 2

**B** 3

**C** 6

**D** 4

**E** Infinitely many

Full Mock Test 2 · Paper 2

# Question 12

A function f permutes {1, 2, 3, 4}. For every x in this set, f(f(x)) = 5 − x. It is also known that f(1) = 2. What is f(3)?

**A** 4

**B** 2

**C** 3

**D** 1

**E** No such permutation exists

Full Mock Test 2 · Paper 2

# Question 13

Each card shows a number and may carry a star, a square, both or neither. Every card with both symbols shows an even number. At least one card shows an odd number and carries a star. Which conclusion follows?

**A** Every card with a star lacks a square.

**B** Some odd-numbered card carries a square.

**C** Some even-numbered card carries both symbols.

**D** Every odd-numbered card carries a star.

**E** Some card with a square lacks a star.

**F** Every even-numbered card carries a square.

**G** Some card with a star lacks a square.

Full Mock Test 2 · Paper 2

# Question 14

For 0 ≤ a ≤ 2 let I(a) be the integral of |x − a| from 0 to 2. Which condition is equivalent to I(a) ≥ 5/4?

**A** |a − 1| ≤ 1/4

**B** |a − 1| ≤ 1/2

**C** |a − 1| ≥ 1/4

**D** |a − 1| ≥ 1/2

**E** |a − 1| ≥ 3/4

Full Mock Test 2 · Paper 2

# Question 15

For a positive integer n, 72 divides $n^2$. What is the smallest possible value of the greatest common divisor of n and 72?

**A** 6

**B** 8

**C** 24

**D** 18

**E** 12

Full Mock Test 2 · Paper 2

# Question 16

For real x and y, consider the pair of conditions |x + y| < 1 and |x − y| < 1. Which single condition is equivalent to the pair?

**A** |x| + |y| < 1

**B** $x^2 + y^2 < 1$

**C** |x| < 1 and |y| < 1

**D** |xy| < 1/4

**E** ||x| − |y|| < 1

Full Mock Test 2 · Paper 2

# Question 17

Joey and Chloe and their daughter Zoe all have the same birthday. Joey is 1 year older than Chloe, and Zoe is exactly 1 year old today. Today is the first of the 9 birthdays on which Chloe's age will be an integral multiple of Zoe's age. What will be the sum of the two digits of Joey's age the next time his age is a multiple of Zoe's age?

**A** 7

**B** 8

**C** 9

**D** 10

**E** 11

Full Mock Test 2 · Paper 2

# Question 18

A token starts at 0. Each move adds either 2 or 5, and two moves of 5 may not be consecutive. In how many different ordered sequences of moves can the token finish at exactly 19?

**A** 5

**B** 7

**C** 9

**D** 8

**E** 10

Full Mock Test 2 · Paper 2

# Question 19

Let x range over [−2, 4] and m over {−1, 2, 4}. Let c be the least radius such that for every x there exists an m with |x − m| ≤ c. Let d be the least radius such that there exists one m with |x − m| ≤ d for every x. What is (c, d)?

**A** (1, 3)

**B** (3/2, 3)

**C** (2, 3)

**D** (3/2, 4)

**E** (2, 4)

Full Mock Test 2 · Paper 2

# Question 20

For which values of k does there exist a quadratic f such that f(0) = f(2) = k, f′(x) > 0 throughout 0 < x < 1, and the integral of f from 0 to 2 is zero?

**A** k = 0

**B** k ≤ 0

**C** k < 0

**D** k > 0

**E** All real k